Yashwanth Boddireddy

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Education

New Jersey Institute of Technology
Master of Science, Data Science, Statistics *Sep 2023 - May 2025*

Work Experience

AI Engineer Jul 2025 - Present
Endeavour Technologies *Jersey City, NJ*

- Created an AI-first workflow using Claude Code to build internal tools: AI writes Python and React code, engineers review on GitHub before shipping, adopted by operations teams daily.
- Built a financial document search system using Python, Pinecone, BM25, Cohere, LangChain, and OpenAI to retrieve and verify answers so analysts spend time on analysis instead of fact-checking.
- Implemented query expansion and multi-hop reasoning to improve retrieval accuracy on complex financial queries, enabling the system to understand nuanced questions across multiple documents.
- Implemented end-to-end LLM application tracing using Langfuse and Python to observe all application calls and identify bottlenecks, so engineers debug issues faster without guessing.
- Built real-time dashboards using Prometheus, Grafana, MySQL, and Elasticsearch to track infrastructure and application health, enabling the team to catch and fix problems instantly.
- Implemented end-to-end ML pipelines using Python and Kubernetes automating data ingestion, feature engineering, model training, packaging, and deployment with Ragas evaluation so models ship without manual bottlenecks.
- Built feature pipelines with validation and versioning to ensure data quality across ML models, preventing model degradation from bad data entering production.

Software Engineer Fellow Jul 2024 - Sep 2024
Headstarter AI *NYC, USA*

- Shipped production AI applications using Python, FastAPI, OpenAI, Pinecone, and Stripe handling real payments and customer data from design to deployment.
- Implemented CI/CD pipelines using Python and GitHub Actions for automated testing, versioning, and reproducible deployments across dev, staging, and production environments.
- Led engineers through full development cycles, reviewing architecture and code quality to ship projects on schedule without technical debt.
- Deployed on AWS with GitHub Actions CI/CD and automated tests to catch problems before users encounter them, ensuring system reliability as requirements changed.

Software Engineer Jan 2023 - Aug 2023
Google *Hyderabad, INDIA*
Google is a client while I was working at Accenture.

- Fixed Bard's (now called Gemini) harmful responses using Python regex patterns to filter political bias, sexual content, hate speech, and misinformation so users got helpful answers.
- Validated content filtering by regenerating responses in production before releases, confirming harmful content was blocked.

Software Engineer Jan 2022 - Aug 2023
Accenture *Hyderabad, INDIA*

- Built NLP classification and ranking models using Python and transformer-based architectures to replace generic recommendations, deployed across apps so users found relevant content.

- Automated ML workflows using Python automating data prep, model training, testing, packaging, and deployment to eliminate manual bottlenecks in release cycles.
- Shipped full-stack features across .NET, C#, React, Angular, and Node.js for enterprise clients, owning the entire path from API design to production.
- Optimized REST endpoints and SQL Server databases using Python and SQL, tuning queries and adding caching so users got instant responses.
- Mentored junior engineers on building and shipping ML systems; two led independent AI projects for clients.

Projects

Multi-Agent Conversational System with Production Concurrency

- Built a customer support system routing conversations to specialists using LangGraph subagents: triage, scheduling, promotions, and support agents run independently for testability and evolution.
- Redesigned concurrency to replace lock-based turn serialization with cancellable, generation-tracked turns, so new utterances cancel stale LLM/tool streams instantly while respecting real constraints like sent SMS.
- Optimized tool-calling by migrating from sequential loops to LangGraph's async ToolNode, running multiple tools in parallel with asyncio.gather to eliminate batching latency.
- Built RAG using LangChain, Chroma, and OpenAI that answers only from the knowledge base and defers honestly, backed by Supabase Postgres with row-level-security for private auditable history.
- Secured the system with Supabase JWT authentication, single-use signed tokens for WebSocket connections, and rate limiting on billable endpoints to prevent abuse.
- Deployed on FastAPI and AWS with React dashboard on Vercel, Shopify integration, and Postgres-backed LangGraph checkpointer. Fixed a PgBouncer bug to restore uptime and enable Stripe, Shopify, and Zendesk integrations.